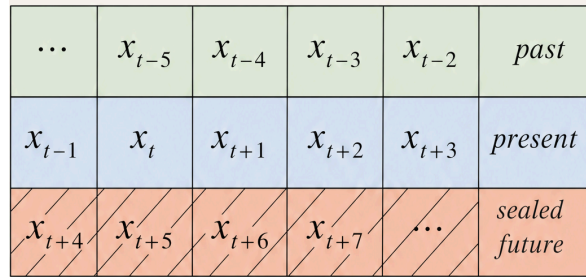
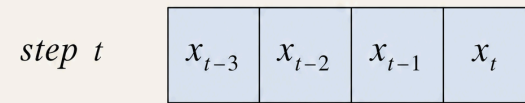
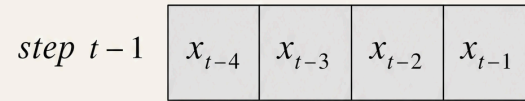
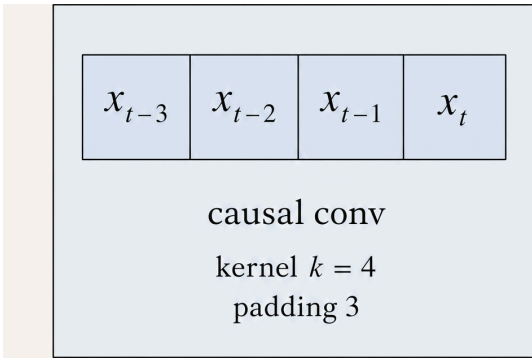
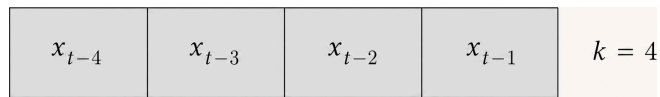
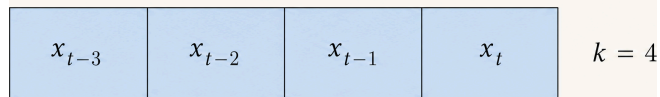


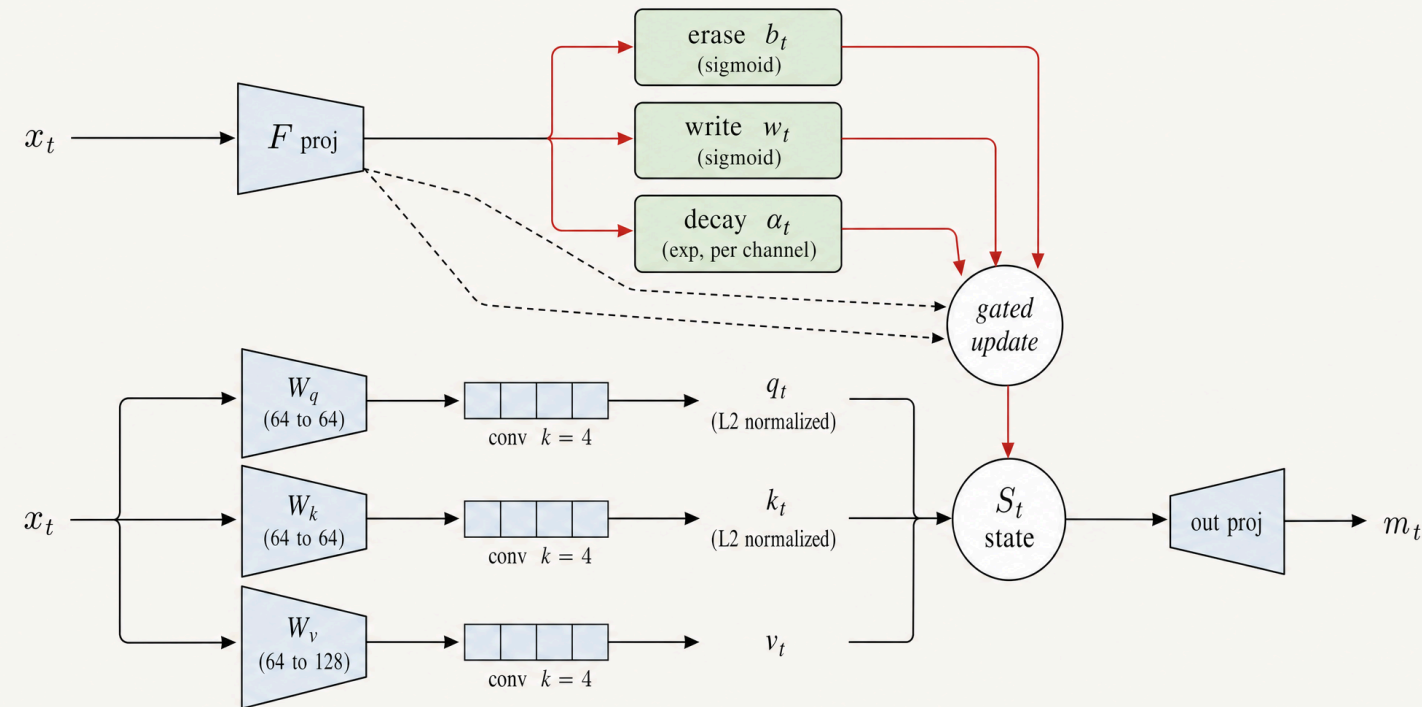
CAUSAL CONVOLUTION WINDOW



kernel $k = 4$ depthwise $\text{groups} = d$ padding $k - 1$

step $t-1$ step t

PROJECTION · GATE GENERATION · MECHANISM



erase

$$S \leftarrow (I - k_t (b_t \odot k_t)^T) S$$

decay

$$S \leftarrow \text{diag}(\alpha_t) S$$

write

$$S \leftarrow S + k_t (w_t \odot v_t)^T$$

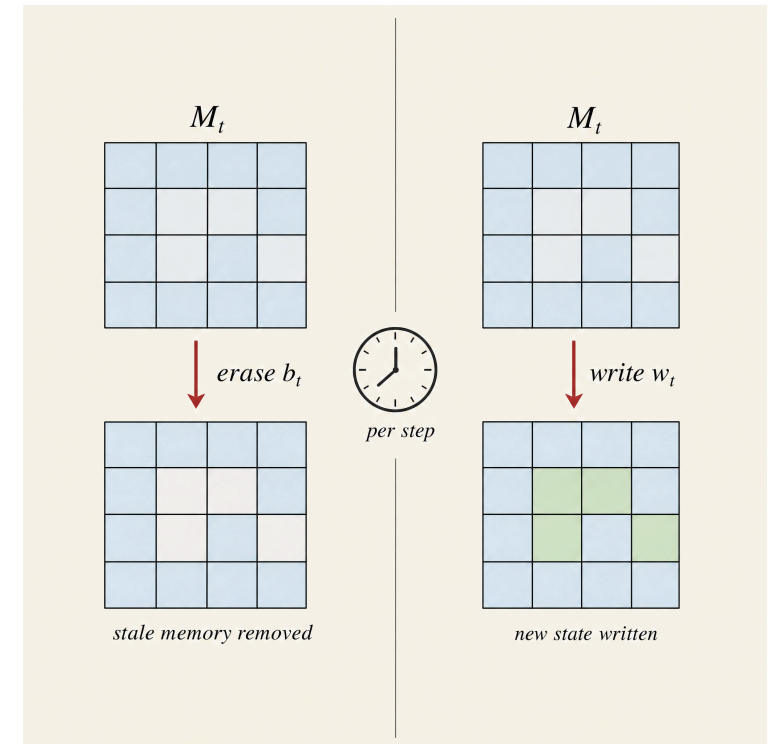
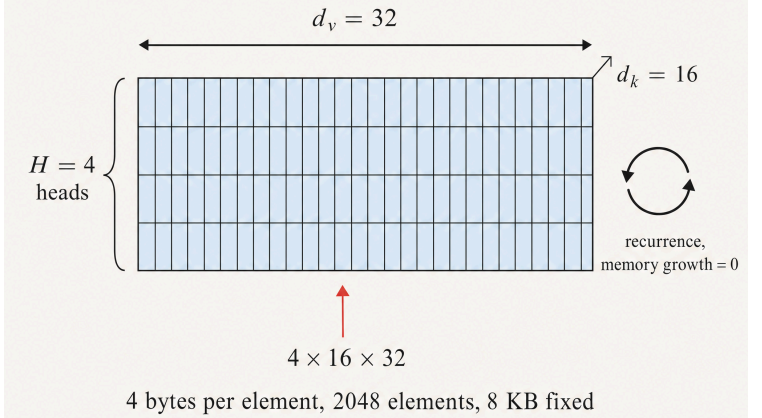
output

$$o_t = S_t^T q_t \rightarrow \text{Linear}(128 \rightarrow 64) \rightarrow \text{RMSNorm}$$

 α law

$$\alpha_t = \exp(g), \quad g = -\exp(A_{\log}) \cdot \text{softplus}(\text{raw}_g + \text{dt_bias})$$

FIXED-SIZE STATE · 8 KB



H = 4

d_k = 16

d_v = 32

state 8 KB

O(L) per layer

per-layer gating